

Yijung Kang

Curriculum Vitae

Rubin Observatory Base Facility
Casilla 603
La Serena, Chile

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Current Position

2022-present **Research Associate & Observing Specialist Postdoc**
Rubin Observatory/SLAC/Stanford/KIPAC, La Serena, Chile
80% for on-site commissioning and operation and 20% for research (first 3 years):
Controlling and operating telescopes & cameras with systems, troubleshooting
Reporting nightly observing log and fault reports
Documentation and training newer team members
Contributing to analysis system performance optimization and verification

Previous Position

2019-2022 **Science Fellow**, Research and Science Services
Gemini Observatory, La Serena, Chile
Postdoctoral fellow, 50% for research and 50% for observatory support:
Queue coordinating, night observation, instrument support, user support

Education

2020 Ph.D. in Astronomy, Yonsei University, Seoul, South Korea
 Advisor : Prof. Young-Wook Lee
 Thesis: “*Evidence for Luminosity Evolution in Type Ia Supernova Cosmology from Spectroscopy of Early-type Host Galaxies*”

2012 M.S. in Astronomy, Yonsei University, Seoul, South Korea
 Advisor : Prof. Young-Wook Lee

2010 B.S. in Astronomy and Physics, Yonsei University, Seoul, South Korea

Language Skills

English (proficient), Spanish (conversant), Korean (native)

Grants, Fellowships, and Telescope Time Awarded

Collaboration

2024 - **LSST Builder: data rights and access to proprietary LSST data**
: Conferred after 2 full-time equivalent years of contributions to the Rubin Observatory Construction and Operations

Successful Telescope Proposals

2024 **Gemini S**
GS-2024B-FT-215 (1.94 hours; PI: **Hyein Yoon**)

2021 **Very Large Array**
VLA/21A-310 (50 hours; PI: **Yijung Kang**)

2015-2016 **Gemini N & S** via K-GMT science program
GS-2015B-Q-41, GN-2016A-Q-51 (total 14 hours; PI: Y.-W. Lee)

2014 **MMT** via K-GMT science program
2014B-UAO-G16 (total 1 night; PI: Y.-W. Lee)

Assistantship & Scholarship

2011-2017 Teaching/Research assistantship, Yonsei University (\$23,460)

2010-2014 BK 21 (Brain Korea 21) & BK 21 Plus Scholarship (academic excellence; \$34,100)

Observational Experiences

Observatory Support

2022-present **Rubin Observatory (~60 nights/year support at the site)**
Day- and night-time commissioning and operating activities at the site
Controlling and operating telescopes, cameras, and systems
Performing basic troubleshooting and evaluation of the delivered image quality
Reporting the nightly observing log and submitting fault reports
Contributing documentations related to observatory operations

2019-2022 **Gemini Observatory**
Preparing for night queue observation plans (~7 weeks/year)
Executing night observations (~15 nights/year)
Specialized program support (~15 queue programs/semester)
Instrument support for Gemini Multi-Object Spectrographs (GMOS)

Optical Long-slit spectroscopy

2015-2016 14 hours at Gemini North & South 8.1 m telescope: GMOS long-slit

2014-2016 10 nights at MMT 6.5 m telescope: Blue Channel Spectrograph

2011-2013 27 nights at Irene du Pont 2.5 m telescope: The Boller & Chiven Spectrograph

2010 8 nights at CTIO SMART 1.5 m telescope: The R-C Spectrograph

Optical Integrated Field Units (IFU) spectroscopy

2011-2012 22 nights at McDonald 2.7 m telescope: Virus-P

Publication List

Refereed Journal Papers

- 8) "Early-type Host Galaxies of Type Ia Supernovae. III. Star Formation History of the Host Galaxies"
Y. Kang et al., in preparation
- 7) "Tracing back the birth environments of Type Ia supernova progenitor stars: a pilot study based on 44 early-type host galaxies"
Y.-L. Kim, L. Galbany, I. Hook, Y. Kang 2024, MNRAS, 529, 3806 [\[ADS\]](#)
- 6) "Evidence for strong progenitor age dependence of type Ia supernova luminosity standardization process"
Y.-W. Lee, C. Chung, P. Demarque, S. Park, J. Son, and Y. Kang 2022, MNRAS, 517, 2697 [\[ADS\]](#)
- 5) "Further Evidence for Significant Luminosity Evolution in Supernova Cosmology"
Y.-W. Lee, C. Chung, Y. Kang, M. J. Jee 2020, ApJ, 903, 22, 22 pages [\[ADS\]](#)
- 4) "Early-type Host Galaxies of Type Ia Supernovae. II. Evidence for Luminosity Evolution in Supernova Cosmology"
Y. Kang, Y.-W. Lee, Y.-L. Kim, et al. 2020, ApJ, 889, 8 [\[ADS\]](#)
- 3) "The Globular Cluster Origin of the Milky Way Outer Bulge: Evidence from Sodium Bimodality"
Y.-W. Lee, J. J. Kim, C. I. Johnson, Y. Kang, et al. 2019, ApJL, 878 [\[ADS\]](#)
- 2) "Environmental Dependence of Type Ia Supernova Luminosities from the YONSEI Supernova Catalog"
Y.-L. Kim, Y. Kang, and Y.-W. Lee 2019, JKAS, 52 [\[ADS\]](#)
- 1) "Early-type Host Galaxies of Type Ia Supernovae. I. Evidence for Downsizing"
Y. Kang, Y.-L. Kim, Dongwook Lim, et al. 2016, ApJS, 223, 7 [\[ADS\]](#)

Proceedings

- 8) "Rubin ToO 2024: Envisioning the Vera C. Rubin Observatory LSST Target of Opportunity program",
I. Andreoni et al. 2024, arXiv e-prints, [\[ADS\]](#) (co-author)
- 7) "Rubin M1M3 support system dynamic performance",
B. C. Quint et al. 2024, SPIE Conference Series, [\[ADS\]](#) (co-author)
- 6) Rubin Observatory Simonyi Survey Telescope integrated mount performance,
B. Stalder et al 2024, SPIE Conference Series [\[ADS\]](#) (co-author)
- 5) The Vera C. Rubin's M2 support system integration and verification at the TMA,
G. Rodeghiero et al. 2024, SPIE Conference Series, [\[ADS\]](#) (co-author)
- 4) Investigation of Stellar Populations in the Early-type Host Galaxies of Type Ia Supernovae
Y. Kang, and Y.-W. Lee 2019, American Astronomical Society Meeting #233, [\[ADS\]](#)
- 3) "Early-type Host Galaxies of Type Ia Supernovae: Origin of the Correlation between Hubble Residual and Host Mass"
Y. Kang, Y.-L. Kim, D. Lim, et al. 2016, American Astronomical Society Meeting #227, [\[ADS\]](#)
- 2) "The Luminosity of Type Ia Supernova and the Properties of Their Early-type Host Galaxies"
Y. Kang, Y.-L. Kim, Y.-W. Lee et al. 2015, PKAS, 30, 487, 2 pages [\[Link\]](#)
- 1) "YONsei Nearby Supernova Evolution Investigation (YONSEI) Supernova Catalogue"
Y.-L. Kim, Y. Kang, Y.-W. Lee 2015, PKAS, 223, 485, 2 pages [\[Link\]](#)